Data selection

Abstract

1 Introduction

The performance of modern machine learning models is determined not only by their architecture and optimization procedure, but critically by the data on which they are trained. This has motivated the *data selection* problem: given a training set $S = \{Z_1, \ldots, Z_n\}$, identify a subset $I \subseteq S$ of size $m \ll n$ such that training on I yields a model with performance *comparable* to training on S.

This problem is variously called coreset selection, dataset pruning, active learning or subset selection depending on the context and the notion of *comparable* performance employed [AHV; Mos+25]. Despite this practical and theoretical importance, a unified framework that both *defines* what a good subset is and *certifies* how close any given selection is to that optimum remains elusive.

Eclectic approaches have been proposed, each optimizing a different proxy for subset quality. Geometry-based methods select a covering subset in feature space, minimizing the maximum distance from any training point to the selected set [AHV; SS18]. Training-oriented methods [Mos+25] score examples using information stemming from their behaviour during learning: forgetting events [Ton+19], gradient norms [PGD21], or self-supervised prototype distances [SGSGM23], pruning low-scoring examples post-hoc. Influence-based approaches [KL20; PLSK20] estimate each example's counterfactual effect on model predictions via first- or second-order approximations to leave-one-out retraining. Gradient matching methods [MBL20; KSRDI21] select subsets whose aggregate gradient closely tracks the full-dataset gradient. Bilevel-optimization approaches [KSRI21] maximize held-out performance as an outer objective, selecting examples most beneficial for a target task. Finally, submodular formulations [WIB15] optimize coverage or diversity functions over gradient or feature similarities via greedy approximation.

A natural question is whether any sample-based procedure can universally identify optimal subsets. To answer this, we distinguish two fundamental obstacles to data selection (Equation (2)). The *compression obstacle* captures the fact that a good full-data model does not guarantee a good size-m subset exists; this sets an irreducible floor on subset risk. The *identification obstacle* is more subtle, and characterizes our problem: even when a perfectly compressible subset exists within the realized sample, the sample alone may not reveal which one it is. Feldman [Fel21] demonstrated that in long-tailed data distributions, rare examples are statistically indistinguishable from useless outliers using only the training data, making it provably impossible to select them correctly without information beyond the sample. This justifies the adoption of data selection methods that optimize performance against a held-out *validation* set [KSRDI21; KSRI21; EFM24], alongside parallel paradigms in data attribution [PLSK20; IPELM22] and data valuation [GZ19] that rely on testing query sets or external samples.

In Section 3, we provide an example describing these obstacles: Theorem 3.1 gives a setting where any data selector using only the training samples underperforms an idealized oracle with direct access to the true population risk function. While this example justifies the usage of samples outside of the training set to perform data selection, we show in **??** that it requires access to *quadratically* more samples in a validation set to be able to match the performance of the data selector we introduce.

These results clarify what is achievable with sample-based data selection and motivate a more modest but well-founded goal. Rather than seeking a selector that universally identifies the best subset, we propose equipping a data selection method with a *certificate*: a computable upper bound on how far the subset-trained model diverges from the full-data model, regardless of whether the optimal subset has been identified. In Section 5, we formalize divergence between the full-data model and a subset model via the Kullback-Leibler (KL) divergence $D_{\mathrm{KL}}(\mathcal{A}_T(S) \parallel \mathcal{A}_T(I))$ between the distributions of model parameters after T training iterations on S and I, respectively. This choice is justified by Proposition 5.1: for *any* sub-Gaussian query function q and *any* randomized learning algorithm, the expected leave-group-out error is bounded by $\sqrt{2\sigma^2 D_{\mathrm{KL}}(\mathcal{A}_T(S) \parallel \mathcal{A}_T(I))}$. The KL divergence thus serves as a universal, task-agnostic proxy for subset quality, upper-bounding the deviation of any downstream "quality", as measured by loss, accuracy, or prediction.

While generally intractable, this divergence becomes tractable when the learning algorithm is the Unadjusted Langevin Algorithm (ULA) [WT11]. We derive a closed-form bound on $D_{\mathrm{KL}}(\mathcal{A}_T(S) \parallel \mathcal{A}_T(I))$ in terms of the trajectory-averaged squared norm of the gradient discrepancy $\|\nabla \Delta_I(\theta_k)\|^2 = \|\nabla L_I(\theta_k) - \nabla L_S(\theta_k)\|^2$ (Theorem 5.2). This bound decomposes over the training examples and can be formalized as a constrained quadratic program (QP) that can be minimized by gradient descent. This framework unifies a broad family of prior data selection methods: FILL HERE WHICH ONES, with the KL bound providing the theoretical certificate that each individually lacked.

2 Preliminaries

2.1 Notation

Let $\mathcal{Z}$ denote the space of data points and $P \in \mathcal{P}(\mathcal{Z})$ denote the underlying data distribution. We consider the setting of empirical risk minimization (ERM) on *i.i.d.* training samples $S = \{Z_1, \ldots, Z_n\} \sim P^n$. A data selection procedure with budget m identifies a subset of indices $I \subseteq [n] = \{1, \ldots, n\}$ with cardinality $|I| \leq m$. Models are parameterized by a vector $\theta \in \mathbb{R}^d$, and the loss incurred on an individual sample Z_i is given by $\ell(\theta, Z_i)$. The full-dataset empirical risk is defined as $L_S(\theta) = \frac{1}{n} \sum_{i=1}^{n} \ell(\theta, Z_i)$, while the corresponding subset empirical risk is denoted by $L_I(\theta) = \frac{1}{m} \sum_{i \in I} l(\theta, Z_i)$. For a given (randomized) learning algorithm $\mathcal{A} : S \to \mathbb{R}^d$, the population risk of a hypothesis $h_I = \mathcal{A}(S_I)$ trained on the subset sequence $S_I = (Z_i)_{i \in I}$ is defined as $R(I) := \mathbb{E}_{Z \sim P}[\ell(h_I, Z)]$, while the optimal population risk over the entire hypothesis class is given by $F^* := \inf_h \mathbb{E}_{Z \sim P}[\ell(h, Z)]$.

To capture the optimization dynamics of randomized learning procedures, we let $\mathcal{A}_k(S)$ and $\mathcal{A}_k(I)$ denote the time-dependent marginal probability distributions of the model parameters $\theta_k \in \mathbb{R}^d$ at iteration k when trained on the full dataset S and the subset I, respectively. At the final training iteration T, these endpoint parameter distributions are denoted by $\mathcal{A}_T(S)$ and $\mathcal{A}_T(I)$. We measure the discrepancy between two probability distributions P, Q by the Kullback-Leibler divergence (KL): $D_{\mathrm{KL}}(P \| Q) := \mathbb{E}_{x \sim P}[\log(P(x)/Q(x))]$.

2.2 Related works

2.3 SGLD

3 Sample-based and validation-based data selection

In this section, we formalize what data selection attempts to achieve and characterize the fundamental information-theoretic limits of three natural selection paradigms.

A natural objective for any data selector is to approximate the best achievable population performance. Denoting $F^\star = \inf_h \mathbb{E}_{Z\sim P}[\ell(h, Z)]$, the quality of a generic selector that returns a subset $\Pi(S) \subseteq [n]$ of the training sample is measured by its excess population risk:

$$R(\Pi(S)) - F^\star. \tag{1}$$

We distinguish between three classes of selectors, ordered by the information they can access. A *sample-only* selector is a (possibly randomized) map $\Pi : S \mapsto I(S) \subseteq [n]$ that produces a subset of the training indices using only S and the learning algorithm $\mathcal{A}$. At the other extreme, an *evaluation-oracle* selector Π_{or} may additionally query the exact population risk $R(I)$ for any candidate subset $I \subseteq [n]$; it serves as an idealized benchmark corresponding to access to unlimited validation data. In practice, exact population risks are unavailable, and one instead uses empirical estimates from a held-out validation set $V = (Z'_1, \ldots, Z'_v) \sim P^v$ [ADD CITATIONS HERE]. This gives rise to a *validation* selector: a map $\Pi_{\text{val}} : (S, V) \mapsto I \subseteq [n]$ that may use $\mathcal{A}$, S, and V jointly. Throughout, we assume that training, validation, and test examples are drawn i.i.d. from the same distribution P.

3.1 A compression–selection decomposition

For a subset budget m, let $R^\star_m(S) := \min_{\substack{I\subseteq[n]\\ |I|\le m}} R(I)$ denote the best achievable population risk over subsets of the training sample of size at most m. The excess risk of any selector then decomposes as

$$R(\Pi(S)) - F^\star = \underbrace{R^\star_m(S) - F^\star}_{\text{compression floor}} + \underbrace{R(I_\Pi(S)) - R^\star_m(S)}_{\text{selection error}}. \tag{2}$$

The compression floor is irreducible: no selector, even one with oracle access, can bring the risk below $R^\star_m(S)$. The selection error captures how well the selector identifies the best available subset. When the selector has direct access to $R(\cdot)$, or to a statistical proxy via a validation set, one might hope to drive this term to zero, recovering $R^\star_m(S)$. The following theorem provides an example where the selection error cannot match an oracle selector unless the selector is given either an exact oracle or a validation set that is quadratically larger than the training data.

3.2 A one-sample oracle separation

To formalize the limitations of minimizing Equation (2) with finite information, we construct an explicit, well-specified noiseless linear-regression family on which the three classes of selectors are separated as sharply as possible. We defer the construction and proofs to Section A.

Let K denote the effective dimension of the problem, tied to the sample size n via the coverage condition $n \geq K \log(12K)$. For any target error $\varepsilon \in (0, 1/8)$, we define a family of distributions $\{P_h : h \in [K]\}$, where h indexes a hidden coordinate. The optimal population risk is $F^\star = 0$, and the learner is the minimum-norm least-squares interpolator, equivalently the limit of gradient descent from the origin on the squared loss.

Theorem 3.1 (One point for the oracle, many for sample-only and validation)**.** *Let n be large enough that there is an integer $K \geq 8$ satisfying $n \geq K \log(12K)$, and take $K = \Theta(n/\log n)$. For every target error $\varepsilon \in (0, 1/8)$, there is a family of well-specified noiseless linear-regression distributions $\{P_h : h \in [K]\}$ and a fixed minimum-norm learner such that the following hold:*

1. ***Full-data training is good.*** *For every $h \in [K]$, there is an event G_h with $\mathbb{P}_{S \sim P_h^n}(G_h) \geq 11/12$ such that, on G_h,*
$$R_h([n]) - F^\star = 0.$$

2. ***An evaluation oracle compresses to one example.*** *On G_h, a distribution-aware evaluation-oracle selector using at most $K - 1$ exact-risk queries outputs a subset $I_{\mathrm{or}}(S)$ of size one satisfying*
$$R_h(I_{\mathrm{or}}(S)) - F^\star \leq \varepsilon.$$

3. ***Sample only selection is bad:*** *For every possibly randomized sample-only selector Π_r that outputs at most $r \leq K/4$ examples, there exists a fixed $h^\star \in [K]$ such that*
$$\mathbb{E}_{S \sim P_{h^\star}^n}\left[R_{h^\star}(I_{\Pi_r}(S)) - F^\star\right] \geq \frac{1}{2}.$$

4. ***Validation selectors are also bad:*** *Let Π be any (possibly randomized) selector that observes the training sample $S \sim P_h^n$ together with an independent hold-out set $V \sim P_h^v$ and outputs a subset $I_\Pi(S, V)$ of at most m training examples, with $1 \leq m < K$. There are constants $c_1, c_2 > 0$ such that if*
$$v \leq c_1 \frac{n^2}{\log n} - c_2 nm,$$
then there exists a fixed $h^\star \in [K]$ for which
$$\mathbb{E}\left[R_{h^\star}(I_\Pi(S, V)) - F^\star\right] \geq \tfrac{1}{2}.$$
Consequently, to output a single example ($m = 1$) and match the oracle, the hold-out set must contain $v = \Omega(n^2/\log n)$ examples, quadratically more than the n training points.

Returning to Equation (2), Theorem 3.1 reveals that the selection error is the crux of the difficulty. On the good event G_h, point 2 shows that the evaluation oracle drives the selection error to zero and finds a near-optimal single training example with excess risk at most ε. Point 3 and 4 establish that this is information-theoretically impossible for any selector lacking oracle access, unless it either selects many examples ($m = \Omega(K)$) or is given a hold-out of size $\Omega(n^2/\log n)$.

The obstruction is one of *identification*. On G_h, the optimal singleton already exists in the realized sample; what the training data alone cannot reveal is which singleton it is. A validation set can in principle resolve the ambiguity, but only at a sample cost that is prohibitive.

Although the construction is adversarial by design, the underlying phenomenon is not pathological. It arises whenever a dataset contains features that are (i) rare relative to the training sample size, (ii) critical

for generalization, and (iii) statistically indistinguishable from irrelevant features using training-sample statistics alone. Maybe give an argument about long-tailed distributions here ?

Limitations of the compression floor as a selection criterion. Beyond the identification problem, the decomposition Equation (2) suffers another drawback as a practical selection criterion: neither term is certifiable. Both the compression floor $R^\star_m(S) - F^\star$ and the selection error $R(I_\Pi(S)) - R^\star_m(S)$ depend on the true population risk $R(\cdot)$, which is inaccessible without the very oracle that Theorem 3.1 shows is indispensable. Furthermore, one needs access to the $R^\star_m$ value to compute the excess error of any data selector using the decomposition of (2).

This motivates an alternative decomposition that anchors to a quantity the learner can actually certify. Provided the full-data algorithm has achieved small population risk $R([n])$, we may write, for any selector Π,

$$R(\Pi(S)) - F^\star \;=\; \underbrace{R([n]) - F^\star}_{\varepsilon_n} \;+\; \underbrace{R(\Pi(S)) - R([n])}_{\text{faithfulness error}}. \tag{3}$$

The first term, ε_n, is the excess error of the original algorithm trained on all of S. It can be bounded by standard learning-theory arguments and, for well-specified problems, decreases with n. It is a certifiable quantity that does not require knowledge of $R^\star_m(S)$. The second term, which we call the *faithfulness error*, measures how much the selector's output diverges from the behavior of the full-data model. A selector with small faithfulness error preserves, in a precise sense, the properties of training on the entire dataset. Wedevelop an information-theoretic framework for bounding and minimizing this term in the next section.

4 Faithful data selection

Data selection is conventionally posed as an optimization problem: from a finite training sample, find the subset that maximizes population performance. As established in Section 3.2, however, identifying this optimal subset from finite data alone is information-theoretically impossible in the presence of rare features, and approximating it via a validation set requires data of quadratically larger size. This unidentifiability motivates a shift in objective. Rather than chasing an optimality that cannot be certified from the data at hand, we seek *faithfulness*: the property that training on the selected subset yields a model that closely tracks the behavior of the full-data model. Unlike optimality, faithfulness is a property a selector can verify and control without oracle access.

We propose a unified information-theoretic framework for data attribution and selection that achieves this. By modeling the training process via the Unadjusted Langevin Algorithm [WT11], we show that minimizing the Kullback–Leibler (KL) divergence between the parameter distributions induced by the subset and the full training set is a sufficient and computable condition for faithfulness.

4.1 Controlling the faithfulness error

The decomposition Equation (3) reduces bounding $R(\Pi(S)) - F^\star$ to bounding the faithfulness error $R(\Pi(S)) - R([n])$. While Theorem 3.1 shows that minimizing the selection error directly is intractable without extensive validation data, controlling faithfulness is a more tractable alternative. Rather than seeking the globally risk-minimizing subset

$$I^\star := \operatorname*{arg\,min}_{I \subseteq S,\ |I| \le m} R(I),$$

we seek the subset that best replicates the behavior of the full-data algorithm. Formally, for any query function $q : \mathbb{R}^d \to \mathbb{R}$, we aim to find and quantify the subset minimizing the leave-group-out error [YBHS24]:

$$I^\star := \arg\min_{I \subseteq S,\ |I| \le m} \left| \mathbb{E}_{\theta \sim \mathcal{A}_T(S)}[q(\theta)] - \mathbb{E}_{\theta' \sim \mathcal{A}_T(I)}[q(\theta')] \right|.$$

This probabilistic formulation is expressive: different choices of q recover a range of objectives, including parameter proximity, prediction consistency, and loss alignment. The following proposition, proved in Section E, shows that the KL divergence between the full-data and subset parameter distributions provides a universal upper bound on the leave-group-out error for any randomized learning algorithm.

Proposition 4.1. *Let $q : \mathbb{R}^d \to \mathbb{R}$ be a query function that is σ-sub-Gaussian under $\mathcal{A}_T(I)$. For any randomized learning algorithm, the leave-group-out error satisfies*

$$\left| \mathbb{E}_{\theta \sim \mathcal{A}_T(S)}[q(\theta)] - \mathbb{E}_{\theta' \sim \mathcal{A}_T(I)}[q(\theta')] \right| \le \sqrt{2\sigma^2 D_{\mathrm{KL}}\big(q\#\mathcal{A}_T(S) \parallel \mathcal{A}_T(I)\big)} \le \sqrt{2\sigma^2 D_{\mathrm{KL}}\big(\mathcal{A}_T(S) \parallel \mathcal{A}_T(I)\big)}. \tag{4}$$

5 Faithful data selection

Data selection is conventionally posed as an optimization problem: from a finite sample, find the subset that maximizes population performance. However, as we establish in Section 3.2, identifying this optimal subset from finite data alone is information-theoretically impossible in the presence of rare features. This unidentifiability motivates a shift in our target. Rather than chasing an optimality we cannot certify, we seek *faithfulness*, which is a property a selector can actually verify from the data it has. We propose a unified information-theoretic framework for data attribution and selection to achieve this. By modeling the training process via the Unadjusted Langevin Algorithm [WT11], we demonstrate that minimizing the Kullback-Leibler (KL) divergence between the subset and full-data parameter distributions is a sufficient, computable condition for this faithfulness.

5.1 Controlling the faithfulness

???? justify our next approach: while validation-based approaches can provide better selection guarantees, there exists instances where one would require a *quadratically* bigger validation set than the training set to do so. Thus, instead of focusing on finding the overall optimal subset (i.e $I^* := \arg\min_{I \subseteq S, |I| \le m} R(I)$, we will aim to find the subset I that better mimics training on the whole training set S. That is, for any query function $q : \mathbb{R} \to \mathbb{R}$, we want to find, and quantify, the subset that minimizes the leave-group-out error [YBHS24]:

$$I^* := \arg\min_{I \subseteq S, |I| \le m} |\mathbb{E}_{\theta \sim \mathcal{A}_T(S)}[q(\theta)] - \mathbb{E}_{\theta' \sim \mathcal{A}_T(I)}[q(\theta')]|.$$

This probabilistic view is expressive. The following proposition, proved in Section E which holds for any randomized learning algorithm, demonstrates that this KL divergence provides a universal upper bound on the leave-group-out error.

Proposition 5.1. *Let $q : \mathbb{R}^d \to \mathbb{R}$ be a query function that is σ-sub gaussian under $\mathcal{A}_T(I)$. For any randomized learning algorithm, the expected leave-group-out error is bounded by:*

$$|\mathbb{E}_{\theta \sim \mathcal{A}_T(S)}[q(\theta)] - \mathbb{E}_{\theta' \sim \mathcal{A}_T(I)}[q(\theta')]| \le \sqrt{2\sigma^2 D_{\mathrm{KL}}(q\#\mathcal{A}_T(S) \| \mathcal{A}_T(I))} \le \sqrt{2\sigma^2 D_{\mathrm{KL}}(\mathcal{A}_T(S) \| \mathcal{A}_T(I))} \tag{5}$$

5.2 Moving to Langevin Dynamics

While the KL divergence serves as an ideal theoretical proxy for attribution, it is generally intractable to compute. In order to obtain a more expressive bound, we restrict our analysis to the Unadjusted Langevin Algorithm (ULA). Under ULA, the parameters are updated via:

$$\theta_{k+1} = \theta_k - h\nabla L_S(\theta_k) + \sqrt{2h/\beta}\,\xi_{k+1} \tag{6}$$

where $h > 0$ is the step size, $\beta > 0$ is the inverse temperature, and $\xi_{k+1} \sim \mathcal{N}(0, I_d)$. By analyzing the path measure of this specific Markov chain, we can bound the endpoint KL divergence using the gradient variance along the optimization trajectory.

Theorem 5.2. *Let $\Delta_I(\theta) := L_I(\theta) - L_S(\theta)$ be the deviation between the subset loss and the full-dataset loss. Let $\mathcal{A}_k(I)$ be the marginal distribution of the ULA iterate at step k when trained on S. Assuming a common initialization μ_0 for both chains, the KL divergence after T iterations satisfies:*

$$D_{\mathrm{KL}}(\mathcal{A}_T(S) \,\|\, \mathcal{A}_T(I)) \le \frac{\beta h}{4} \sum_{k=0}^{T-1} \mathbb{E}_{\theta_k \sim \mathcal{A}_k(S)} \left[\|\nabla \Delta_I(\theta_k)\|^2 \right] \tag{7}$$

The term $\|\nabla \Delta_I(\theta_k)\|^2$ represents the squared norm of the difference between the subset gradient and the full-batch gradient. By expanding this squared norm, the combinatorial search for the optimal subset that minimizes this upper bound can be exactly reformulated as an integer quadratic program.

Theorem 5.3. *Let the subset $I \subseteq S$ of size m be represented by the indicator vector $\alpha \in \{0,1\}^n$, where $\alpha_i = \mathbb{I}(Z_i \in I)$. Minimizing the KL divergence upper bound established in Theorem 5.2 with respect to I is equivalent to solving the following integer quadratic program:*

$$\min_{\alpha \in \{0,1\}^n} \frac{1}{m^2} \alpha^\top P \alpha + \alpha^\top q \tag{8}$$

subject to the cardinality constraint $\mathbf{1}_n^\top \alpha = m$, where the positive semi-definite Gram matrix $P \in \mathbb{R}^{n \times n}$ and the alignment vector $q \in \mathbb{R}^n$ are given by:

$$P_{ij} = \sum_{k=0}^{T-1} \mathbb{E}_{\theta_k \sim \mathcal{A}_k(S)} \left[\langle \nabla \ell(\theta_k, Z_i), \nabla \ell(\theta_k, Z_j) \rangle \right] \tag{9}$$

$$q_i = -\frac{2}{m} \sum_{k=0}^{T-1} \mathbb{E}_{\theta_k \sim \mathcal{A}_k(S)} \left[\langle \nabla \ell(\theta_k, Z_i), \nabla L_S(\theta_k) \rangle \right] \tag{10}$$

Note: The problem can also be written entirely with respect to P. We have:

$$(5.3) \iff \min_{\alpha \in \{0,1\}^n, \sum_{i=1}^n \alpha_i \le m} \frac{1}{m^2} \alpha^\top P \alpha - \frac{2}{mn} \alpha^\top P \mathbf{1}$$

If the gradients are orthogonal, this reduces to:

$$\begin{aligned} \|\nabla L_I(\theta) - \nabla L_S(\theta)\|^2 &= \|\nabla L_I(\theta)\|^2 - 2\langle \nabla L_I(\theta), \nabla L_S(\theta) \rangle + \|L_S(\theta)\|^2 \\ &= \|L_I(\theta)\|^2 - \frac{2}{n} \|\nabla L_I(\theta)\|^2 + \|L_S(\theta)\|^2 \\ &= \left(\frac{1}{m^2} - \frac{2}{mn}\right) \sum_{i \in I} \|\ell(\theta, x_i)\|^2 + \|L_S(\theta)\|^2 \end{aligned}$$

The Gram matrix P requires computing $\Omega(n^2)$ inner products for T iterations, and thus this can prove to be intractable even for moderately large datasets. In the next part, we will provide an **unbiased** estimator based on *SGLD* (Stochastic Gradient Langevin Dynamics), the mini-batch counterpart of ULA.

5.3 Estimating the Gram Matrix

We define an iteration of Stochastic Gradient Langevin Dynamics (SGLD) as the mini-batch counterpart of Equation (6): Given current iterate θ_k and a mini-batch $B_k \subset [n]$ sampled uniformly at random from $\mathcal{B}$, the next iterate θ_{k+1} is given by:

$$\theta_{k+1} = \theta_k - h\nabla\tilde{L}_{B_k}(\theta_k) + \sqrt{\frac{2h}{\beta}}\xi_k, \quad \xi_k \sim \mathcal{N}(0, I_d), \tag{11}$$

where $\nabla\tilde{L}_{B_k}(\theta) := \frac{1}{|B_k|}\sum_{i\in B_k}\nabla\ell(\theta, Z_i)$ denotes the stochastic gradient evaluated on the mini-batch B_k, and $h > 0$ is the step size (learning rate).

For notation clarity, we denote by P^{ULA} the Gram matrix derived in Equation (9) using (full batch) ULA, and we denote the mini-batch importance sampling estimate as:

$$\hat{P}_{ij}^{\text{SGLD}} = \sum_{k=1}^{T} W_{0:k} \cdot \frac{\mathbb{1}[i \in B_k,\ j \in B_k]}{\pi_{ij}} \big\langle \nabla\ell(\theta_k, Z_i),\ \nabla\ell(\theta_k, Z_j)\big\rangle.$$

where $W_{0:k} = \prod_{j=1}^{k} w_j$, $w_j = \frac{p^{\text{ULA}}(\theta_{k+1}|\theta_k)}{p^{\text{SGLD}}(\theta_{k+1}|\theta_k, B_k)}$ define the per-step and cumulative importance sampling weights, and $\pi_{ij} = \mathbb{P}(Z_i \in B_k, Z_j \in B_k)$. In the theorem below, we establish that $\hat{P}$ is an unbiased estimator of the ULA gram matrix P.

Theorem 5.4 (Unbiasedness of the Path IS Estimator)**.** *For all $i, j \in [n]$:*

$$\mathbb{E}^{\text{SGLD}}\Big[\hat{P}_{ij}^{\text{path}}\Big] = P_{ij}^{\text{ULA}} := \sum_{k=0}^{T-1} \mathbb{E}_{\theta_k\sim\mathcal{A}_k(S)}\big[\big\langle \nabla\ell(\theta_k, Z_i),\ \nabla\ell(\theta_k, Z_j)\big\rangle\big].$$

Proof in appendix

5.4 Another idea: Stochastic estimation on the optimization trajectory

Thank you Mahdi for this amazing idea !!
Storing the exact matrix P requires $O(n \times n)$ storage, which becomes intractable even for moderately large datasets (13 terabytes for ImageNet). Our optimization formulation in Theorem 5.3, if solved by gradient descent, does not require to have access to P directly. Indeed, the gradient update for α can be written as follows:

$$\alpha_{k+1} = \alpha_k - \eta_k(P\alpha_k + q)$$

Therefore, we are equally happy if we can directly compute $P\alpha_k$ (storage of $O(n)$ without storing / computing P. Since $\alpha \in [0, 1]^n$, we have that:

$$P\alpha = \sum_{k=1}^{T} G_k^\top G_k \alpha$$

Consider the following sampling procedure. Let $\xi \in \{0,1\}^n$such that $\xi_i \sim \text{Ber}(\alpha_i)$. Then, defining $\widehat{P\alpha} := \sum_{k=0}^{T} G_k^\top G_k \xi$, we have that:

$$\begin{aligned}\mathbb{E}_\xi\left[\widehat{P\alpha}\right] &= \sum_{k=0}^{T} G_k^\top G_k \mathbb{E}[\xi] \\ &= P\alpha\end{aligned}$$

[TODO: add the variance computation. I have it written in my draft, and is upper bounded by TL^4m]. Note that $G_k\xi = \sum_{i:\xi_i=1} \nabla\ell(\theta_k, Z_i)$, which requires "only" $m \times d$ storage. However, we still have a problem of having to compute and store $G_k^\top$, that is an $n \times d$ matrix, which could be expensive even for moderately large models and datasets. The following suggestion reduces this further:

At step l of the optimization of α, draw ξ as described above, (which gives a set of sampled points I_l, and draw also independently of ξ a subset $J_t \subset [n]$ (uniformly without replacement), such that $|J_l| = b$. Define the new estimator as:

$$(\widetilde{\nabla_\alpha})_j = \frac{n}{b}\mathbb{1}(j \in J_l)\left(\frac{2}{m^2}\widehat{P\alpha}_j + q_j\right), j = 1, \ldots, n;$$

where $\widehat{P\alpha}$ use I_l. This is also an unbiased estimator, since:

$$\begin{aligned}\mathbb{E}_{J_l,I_l}\left[(\widetilde{\nabla_\alpha})_j\right] &= \mathbb{E}_{J_l,I_l}\left[\frac{n}{b}\mathbb{1}(j \in J_l)\left(\frac{2}{m^2}\widehat{P\alpha}_j + q_j\right)\right] \\ &= \mathbb{E}_{J_l}\left[\mathbb{E}_{I_l}\left[\frac{n}{b}\mathbb{1}(j \in J_l)\left(\frac{2}{m^2}\widehat{P\alpha}_j + q_j\right)|J_l\right]\right] \\ &= \mathbb{E}_{J_l}\left[\frac{n}{b}\mathbb{1}(j \in J_l)((P\alpha)_j + q_j)\right] \\ &= (P\alpha)_j + q_j\end{aligned}$$

[TODO] compute variance of this estimator

This yields the following algorithm:

Algorithm 1 Checkpoint-Based Optimization Loop

Precompute once:

Cache $\theta_0, \ldots, \theta_{T-1}$ and compute $\bar{g}_k = \nabla L_S(\theta_k)$ for all k

Compute $q_i = -\frac{2}{m}\sum_k \langle g_i^k, \bar{g}_k\rangle$ for all i ▷ $O(nT)$ backward passes, $O(n)$ storage

for $t = 1, \ldots, T_{\text{outer}}$ **do**

  Draw $\hat{I}_t \sim$ Poisson-sampling(α_t)

  Draw $J_t \sim$ Uniform($[n]$, size b)

  for each checkpoint k **do** ▷ resample $\hat{I}_t$ fresh per k, might reduce variance

    $\hat{u}_t^k = \sum_{i\in\hat{I}_t} \nabla\ell(\theta_k, Z_i)$ ▷ $O(m)$ backward passes

    for $j \in J_t$ **do**

      Accumulate $\langle g_j^k, \hat{u}_t^k\rangle$ into $\hat{P}_{\alpha_t}[j]$ ▷ $O(b)$ backward passes

  Compute $\hat{g}_t[j] = \begin{cases} \frac{n}{b}\left(\frac{2}{m^2}\hat{P}_{\alpha_t}[j] + q_j\right) & \text{for } j \in J_t \\ 0 & \text{otherwise} \end{cases}$

  $\alpha_{t+1} = \Pi_C[\alpha_t - \eta_t\hat{g}_t]$ where $C = \{\alpha \in [0,1]^n | \mathbf{1}^\top\alpha = m\}$

The issue that we face here is as follows. Recall the optimization problem: $\min_{\alpha\in\{0,1\}^n} \frac{1}{m^2}\alpha^\top P\alpha + \alpha^\top q$, such that $\mathbf{1}^\top\alpha = m$. In order to use first order optimization methods, we relax the formulation so that $\alpha \in [0,1]^n, \mathbf{1}^\top\alpha = m$. Consider the following point $\alpha_p = \frac{m}{n}\mathbf{1}$. This point is feasible with respect to the constraints, and the gradient of the objective function evaluated at α_p is zero:

$$\nabla\left(\frac{1}{m^2}\alpha^\top P\alpha + \alpha^\top q\right) = \frac{1}{m^2}P\alpha + q$$

and by plugging the value of α_p this is zero. Therefore, α_p is a global minimizer of the problem (since it is a convex problem), yet it does not give anything interesting in terms of data selection (all points have an equal weight). I thought of adding regularization but I'm afraid this will give us a very sparse α and we won't be able to sample too much points for estimating $P\alpha$.

5.5 Another technique based on pipage rounding

Thank you Mahdi again for this idea !

Let $\mathcal{B}_m := \{\alpha \in \{0,1\}^n : 1^\top\alpha = m\}$ and consider the objective

$$f(\alpha) := \frac{1}{m^2}\alpha^\top P\alpha + q^\top\alpha, \quad 1 \le m \le n-1,\ P \succeq 0.$$

For i with $\alpha_i = 1$ and j with $\alpha_j = 0$, the change produced by exchanging i and j is defined as:

$$\Delta_{ij}(\alpha) := f(\alpha + e_j - e_i) - f(\alpha) = \frac{2}{m^2}\big((P\alpha)_j - (P\alpha)_i\big) + q_j - q_i + \frac{P_{ii} + P_{jj} - 2P_{ij}}{m^2}.$$

Algorithm 2 Corrected swap-based subset selection

1: Initialize $\alpha_0 \in \mathcal{B}_m$, and set $t = 0$.
2: Choose
$$(i_t, j_t) \in \arg\min_{\substack{i:\alpha_{t,i}=1\\ j:\alpha_{t,j}=0}} \Delta_{ij}(\alpha_t)$$
3: If $\Delta_{i_t j_t}(\alpha_t) < 0$, set $\alpha_{t+1} := \alpha_t + e_{j_t} - e_{i_t}$, increment t, and return to Step 2. Otherwise, return α_t.

$$q = \frac{2}{mn}P\mathbf{1}$$

Every iterate remains in $\mathcal{B}_m$, and since every accepted swap strictly decreases f, the algorithm terminates in at most $\binom{n}{m} - 1$ iterations.

Because $P \succeq 0$, fix vectors $\phi_1, \dots, \phi_n$ satisfying $P_{ij} = \langle\phi_i, \phi_j\rangle$. Define

$$r := \min\{m, n-m\}, \quad \kappa := \frac{1}{m^2}\max_{i,j}(e_i - e_j)^\top P(e_i - e_j) = \frac{1}{m^2}\max_{i,j}\|\phi_i - \phi_j\|^2. \tag{3}$$

Let $\alpha^* \in \arg\min_{\alpha\in\mathcal{B}_m} f(\alpha)$ and $H_t := f(\alpha_t) - f(\alpha^*)$.

Theorem 5.5 (Convergence and excess objective). *Let α_t be the iterate produced by corrected Procedure 2, and after termination extend the sequence by setting $\alpha_t = \alpha_{\mathrm{out}}$. Then, for every $t \geq 0$,*

$$H_t \leq r\kappa + \left(1 - \frac{1}{r}\right)^t [H_0 - r\kappa]_+. \tag{4}$$

Consequently, the returned solution satisfies

$$\begin{aligned} f(\alpha_{\mathrm{out}}) - f(\alpha^*) &\leq r\kappa \\ &= \frac{\min\{m, n-m\}}{m^2} \max_{i,j} (e_i - e_j)^\top P (e_i - e_j) \\ &= \frac{\min\{m, n-m\}}{m^2} \max_{i,j} \|\phi_i - \phi_j\|^2. \end{aligned} \tag{5}$$

In particular, when $m \leq n/2$,

$$f(\alpha_{\mathrm{out}}) - f(\alpha^*) \leq \frac{1}{m} \max_{i,j} \|\phi_i - \phi_j\|^2.$$

For every $\varepsilon > 0$, the bound $H_t \leq r\kappa + \varepsilon$ holds after at most

$$\left\lceil r \log\left(\frac{[H_0 - r\kappa]_+}{\varepsilon}\right)\right\rceil_+$$

accepted swaps, with the displayed quantity interpreted as zero when its numerator is zero, and where $\lceil x \rceil_+ := \max\{0, \lceil x \rceil\}$.

Proof. Fix an iterate $\alpha = \alpha_t$. Let

$$A := \{i : \alpha_i = 1, \alpha_i^* = 0\}, \quad B := \{j : \alpha_j = 0, \alpha_j^* = 1\}.$$

The two sets have the same cardinality $k := |A| = |B| \leq r$. If $k = 0$, then $\alpha = \alpha^*$ and there is nothing to prove. Otherwise, pair their elements as $(i_1, j_1), \ldots, (i_k, j_k)$ and define

$$d_l := e_{j_l} - e_{i_l}, \quad D := \sum_{l=1}^{k} d_l = \alpha^* - \alpha.$$

By (1), the change associated with the l-th exchange is

$$\Delta_{i_l j_l}(\alpha) = \frac{2}{m^2} \alpha^\top P d_l + q^\top d_l + \frac{1}{m^2} d_l^\top P d_l.$$

On the other hand, since $\alpha^* = \alpha + D$,

$$-H_t = f(\alpha^*) - f(\alpha) = \frac{2}{m^2} \alpha^\top P D + q^\top D + \frac{1}{m^2} D^\top P D.$$

Summing the k swap changes and using $P \geq 0$ therefore gives

$$\sum_{l=1}^{k} \Delta_{i_l j_l}(\alpha) = -H_t + \frac{1}{m^2} \left(\sum_{l=1}^{k} d_l^\top P d_l - D^\top P D \right) \leq -H_t + k\kappa. \tag{6}$$

Hence at least one feasible exchange satisfies

$$\Delta_{ij}(\alpha_t) \le -\frac{H_t}{k} + \kappa \le -\frac{H_t}{r} + \kappa = -\frac{H_t - r\kappa}{r}.$$

Whenever $H_t > r\kappa$, the right-hand side is negative. Because Procedure 2 chooses the best exchange, it cannot terminate and

$$H_{t+1} - r\kappa \le \left(1 - \frac{1}{r}\right)(H_t - r\kappa).$$

If $H_t \le r\kappa$, monotonicity ensures that every later iterate has the same property. Iterating the preceding contraction proves (4).

At termination every feasible exchange has nonnegative change. If $H_{\text{out}} > r\kappa$, the preceding argument would produce a strictly improving exchange, a contradiction. Thus $H_{\text{out}} \le r\kappa$, proving (5). Finally, $(1 - 1/r)^t \le e^{-t/r}$ yields the stated iteration bound. □

How complex is solving the inner loop ?

5.6 Better analysis and scalable variants

We next analyze a matrix-free variant of Procedure 2. It ranks exchanges by the gradient, evaluates only the best-ranked candidate, and preserves integrality by one-for-one exchanges.

Centering and normalization. Let P_0 denote the Gram matrix in the quadratic program above, let $b := -2P_0\mathbf{1}/(mn)$ denote its linear coefficient (called q above), and factor $(P_0)_{ij} = \langle \phi_i, \phi_j \rangle$. With

$$\bar{\phi} := \frac{1}{n}\sum_{i=1}^{n} \phi_i, \qquad z_i := \phi_i - \bar{\phi}, \qquad (P_c)_{ij} := \langle z_i, z_j \rangle,$$

we have $\sum_i z_i = 0$ and, on the Boolean cardinality slice

$$\mathcal{B}_m = \{\alpha \in \{0,1\}^n : \mathbf{1}^\top \alpha = m\}.$$

the discrepancy is

$$D(\alpha) := \left\| \frac{1}{m}\sum_{i=1}^{n} \alpha_i \phi_i - \bar{\phi} \right\|^2 = \frac{1}{m^2}\alpha^\top P_c \alpha = f(\alpha) + \|\bar{\phi}\|^2. \tag{12}$$

Thus D and the objective of Procedure 2 have identical excesses and exchange differences. For the remainder of this subsection, we write $P := P_c$, so

$$D(\alpha) = \frac{1}{m^2}\alpha^\top P \alpha, \qquad h(\alpha) := \nabla D(\alpha) = \frac{2}{m^2} P\alpha, \qquad \succeq 0.$$

Equivalently, one may retain an uncentered representation $f(\alpha) = m^{-2}\alpha^\top P_0 \alpha + b^\top \alpha + c$ and use $h(\alpha) = 2P_0\alpha/m^2 + b$; all gradient-based identities below are unchanged. Fix

$$\alpha^\star \in \operatorname*{arg\,min}_{\alpha \in \mathcal{A}_m} D(\alpha), \qquad D^\star := D(\alpha^\star), \qquad H_t := D(\alpha_t) - D^\star, \qquad r := \min\{m, n-m\},$$

and write $[x]_+ := \max\{x, 0\}$.

Suppose $h_t := h(\alpha_t)$ is stored. The pair

$$i_t \in \arg\max_{i:\alpha_{t,i}=1} h_{t,i}, \qquad j_t \in \arg\min_{j:\alpha_{t,j}=0} h_{t,j}$$

minimizes the linearized change $h_{t,j} - h_{t,i}$. Define $\gamma_t := h_{t,i_t} - h_{t,j_t}$ and $\delta_t := e_{j_t} - e_{i_t}$.

Algorithm 3 Gradient-ranked single-swap descent

Draw $\alpha_0 \sim \mathrm{Unif}(\mathcal{B}_m)$ and query $P\alpha_0$ to store h_0.
Compute $D(\alpha_0)$ from $P\alpha_0$ and set $\underline{D}_{-1} := -\infty$.
for $t = 0, 1, \ldots$ **do**
  Choose i_t, j_t, γ_t, and δ_t as above.
  Set $\underline{D}_t := \max\{\underline{D}_{t-1}, D(\alpha_t) - r[\gamma_t]_+\}$ and $\widehat{H}_t := D(\alpha_t) - \underline{D}_t$.
  if $\gamma_t \le 0$ **then**
    return $(\alpha_t, \widehat{H}_t)$.
  Set $\widetilde{\alpha}_t := \alpha_t + \delta_t$, query $P\widetilde{\alpha}_t$, and store $\widetilde{h}_t := h(\widetilde{\alpha}_t)$.
  Compute $\Delta_t := \frac{1}{2}\delta_t^\top(h_t + \widetilde{h}_t)$.
  if $\Delta_t < 0$ **then**
    Set $(\alpha_{t+1}, h_{t+1}) := (\widetilde{\alpha}_t, \widetilde{h}_t)$ and store $D(\alpha_{t+1}) := D(\alpha_t) + \Delta_t$.
  else
    return $(\alpha_t, \widehat{H}_t)$.

Lemma 5.6 (Exact change and gradient-gap certificate)**.** *Whenever the candidate is tested, its exact objective change is*

$$\Delta_t = -\gamma_t + \frac{1}{m^2}\delta_t^\top P\delta_t. \tag{13}$$

Moreover, at every iterate,

$$H_t = D(\alpha_t) - D^\star \le r[\gamma_t]_+. \tag{14}$$

In particular, the right-hand side is a computable certificate of suboptimality, and $\gamma_t \le 0$ implies that α_t is globally optimal.

Proof. For any feasible $\alpha, \alpha + \delta$,

$$D(\alpha + \delta) - D(\alpha) = \frac{1}{2}\delta^\top\big(h(\alpha) + h(\alpha + \delta)\big) = \delta^\top h(\alpha) + \frac{1}{m^2}\delta^\top P\delta.$$

Since $\delta_t^\top h_t = -\gamma_t$, this proves (13).

Fix an optimum $\alpha^\star$. Relative to $\alpha^\star$, the iterate α_t contains $k \le r$ extra indices $i_1, \ldots, i_k$ and misses k indices $j_1, \ldots, j_k$. By convexity,

$$H_t \le h_t^\top(\alpha_t - \alpha^\star) = \sum_{\ell=1}^{k}(h_{t,i_\ell} - h_{t,j_\ell}) \le k\gamma_t \le r\gamma_t.$$

If $k = 0$, then $H_t = 0$; otherwise the last display and $H_t \ge 0$ imply $\gamma_t \ge 0$. This proves (14). □

Corollary 5.7 (Monotone certified excess)**.** *Every iterate supplies the lower bound*

$$D^\star \ge D(\alpha_t) - r[\gamma_t]_+.$$

Define the best lower bound observed through iteration t and its associated certified excess by

$$\underline{D}_t := \max_{0\le s\le t}\{D(\alpha_s) - r[\gamma_s]_+\}, \qquad \widehat{H}_t := D(\alpha_t) - \underline{D}_t. \tag{15}$$

Then

$$H_t \le \widehat{H}_t, \qquad \widehat{H}_{t+1} \le \widehat{H}_t \tag{16}$$

whenever iteration $t+1$ is reached. Equivalently,

$$H_t \le \widehat{H}_t = \min_{0\le s\le t}\left\{r[\gamma_s]_+ - \big(D(\alpha_s) - D(\alpha_t)\big)\right\} \le r \min_{0\le s\le t}[\gamma_s]_+. \tag{17}$$

Proof. The lower bound follows from Lemma 5.6; hence $\underline{D}_t \le D^\star \le D(\alpha_t)$, which proves $0 \le H_t \le \widehat{H}_t$. If iteration $t+1$ is reached, then $D(\alpha_{t+1}) < D(\alpha_t)$, whereas $\underline{D}_{t+1} \ge \underline{D}_t$, proving monotonicity. Finally, substituting the definition of $\underline{D}_t$ gives the equality in (17); its last inequality uses $D(\alpha_s) \ge D(\alpha_t)$ for every $s \le t$. □

Theorem 5.8 (Suboptimality of the scalable swap)**.** *Let T be the final iteration of algorithm 3, let $\alpha_{\text{out}} := \alpha_T$, and let γ_{out} and δ_{out} denote the gradient gap and exchange direction at that iteration. The algorithm returns the valid certificate $\widehat{H}_T$, with*

$$D(\alpha_{\text{out}}) - D^\star = f(\alpha_{\text{out}}) - \min_{\alpha\in\mathcal{A}_m} f(\alpha) \le \widehat{H}_T \le r \min_{0\le s\le T}[\gamma_s]_+ \le r[\gamma_{\text{out}}]_+. \tag{18}$$

More precisely, if $\gamma_{\text{out}} \le 0$, then $D(\alpha_{\text{out}}) = D^\star$. Otherwise, the tested candidate fails the exact check and

$$0 \le D(\alpha_{\text{out}}) - D^\star \le r\gamma_{\text{out}} \le \frac{r}{m^2}\delta_{\text{out}}^\top P\delta_{\text{out}}. \tag{19}$$

Thus $\widehat{H}_T$ is directly observable and is never weaker than the last-iterate certificate $r[\gamma_{\text{out}}]_+$.

Proof. Every accepted swap strictly decreases D, so the algorithm terminates because $\mathcal{B}_m$ is finite. The bound (18) follows from Corollary 5.7. If the final candidate is rejected, then $\Delta_{\text{out}} \ge 0$, and (13) gives $\gamma_{\text{out}} \le \delta_{\text{out}}^\top P\delta_{\text{out}}/m^2$, proving (19). □

Remark 5.9 (Why history helps). The gaps γ_t need not decrease: an exchange changes both the gradient and the selected and unselected index sets. In contrast, $\widehat{H}_t$ decreases monotonically. Since the choice $s = t$ appears in (17), $\widehat{H}_t \le r[\gamma_t]_+$, so the history-aware certificate is always at least as tight as the certificate from the last iterate. It can also be strictly tighter than $r\min_{s\le t}[\gamma_s]_+$, because it subtracts the objective improvement $D(\alpha_s) - D(\alpha_t)$ made after an earlier certificate was computed. Maintaining it requires only two additional scalars and no additional P-matvecs. ◁

Remark 5.10 (Cost and limitation). The initial gradient costs one P-matvec, and each attempted swap costs one additional P-matvec. The indices i_t and j_t are found by two linear scans, so the remaining work is $O(n)$ per attempt and the working memory is $O(n)$. The method neither forms P, accesses $P_{i_t j_t}$, nor enumerates the $m(n-m)$ feasible pairs. Failure of the tested candidate does not imply one-swap local optimality: another pair may have a smaller quadratic curvature and still decrease D. Tracking $\underline{D}_t$ and $\widehat{H}_t$ adds only $O(1)$ arithmetic and memory per iteration. ◁

5.7 Comparison against baselines

Comparing the bound with random selection: Let U denote the uniform distribution over $\mathcal{B}_m$. We are interested in the following quantity:

$$\mathbb{E}_{\alpha\sim U}[f(\alpha)] = \mathbb{E}_U\left[\alpha^T P\alpha + \alpha^T q\right]$$

Since $\mathbb{E}_U[\alpha] = \frac{m}{n}\mathbf{1}_n$, then we have using linearity of expectation:

$$\mathbb{E}_{\alpha\sim U}[f(\alpha)] = \sum_i \mathbb{E}_U[\alpha_i^2 P_{ii}] + \sum_{i\neq j}\mathbb{E}_U[\alpha_i\alpha_j P_{ij}] + \frac{m}{n}\mathbf{1}_n^T q$$

We use the following quick results on α's moments:

- Since $\alpha_i \in \{0,1\}$, $\alpha_i^2 = \alpha_i$ therefore $\mathbb{E}_U[\alpha_i^2] = \mathbb{E}_U[\alpha_i] = \frac{m}{n}$.
- for $i \neq j$, $\mathbb{E}_U[\alpha_i\alpha_j] = \mathbb{P}_U(\alpha_i = 1, \alpha_j = 1) = \frac{m(m-1)}{n(n-1)}$.
- $q = -\frac{2}{mn}P\mathbf{1}$

Thus, we have:

$$\begin{aligned}
\mathbb{E}_{\alpha\sim U}[f(\alpha)] &= tr(P)\frac{m}{n} + \frac{m(m-1)}{n(n-1)}(1^\top P1 - tr(P)) - \frac{2}{n^2}\mathbf{1}^T P1 \\
&= tr(P)\left[\frac{m}{n} - \frac{m(m-1)}{n(n-1)}\right] + \mathbf{1}^T P1\left[\frac{m(m-1)}{n(n-1)} - \frac{2}{n^2}\right] \\
&= \frac{m(n-m)}{n(n-1)}\mathrm{tr}(P) + \left[\left(\frac{m^2-2}{n^2}\right) - \frac{m(n-m)}{n^2(n-1)}\right]\mathbf{1}^T P\mathbf{1} \\
&= \frac{m(n-m)}{n(n-1)}\mathrm{tr}(P) - \frac{m(n-m)}{n^2(n-1)}\mathbf{1}^T P\mathbf{1} + f\left(\frac{m}{n}\mathbf{1}\right) \\
&= \frac{m(n-m)}{n^2(n-1)}\Big(n\times tr(P) - \mathbf{1}^T P1\Big).
\end{aligned}$$

5.8 Connection to existing methods for data selection

6 Experiments

A Construction of the example and proof of Theorem 3.1

We give a well-specified noiseless linear-regression construction. The full sample is good, a distribution-aware evaluation oracle can compress to one training example, but every sample-only selector needs many more examples to achieve comparable population error.

The parameter K is the effective dimension, i.e. the number of plausible coordinates among which the useful one is hidden. It is tied to the sample size through the coverage condition

$$n \geq K\log(12K).$$

The theorem below takes the largest feasible K, which gives $K = \Theta(n/\log n)$. For the construction, fix any such K, a target error $\varepsilon \in (0, 1/8)$, and a hidden coordinate $h \in [K]$. Set

$$\rho := \frac{1}{16n}, \qquad a^2 := \varepsilon, \qquad A^2 := \varepsilon + \frac{K}{\rho}.$$

The distribution P_h is defined as follows. Draw $U \sim \mathrm{Unif}([K])$. If $U = h$, set $T = A$ with probability ρ and $T = a$ otherwise. If $U \neq h$, set $T = a$. Finally set

$$X = Te_U, \qquad Y = T.$$

Training and test examples are drawn i.i.d. from the same P_h. The model is well-specified with $\theta^\star = \mathbf{1}_K$ and $F^\star = 0$.

Let $\mathcal{A}$ be gradient descent from the origin on squared loss, run to convergence. Equivalently, $\mathcal{A}$ returns the minimum-norm least-squares interpolator. If $C_I \subseteq [K]$ is the set of coordinates appearing in S_I, then

$$\widehat{\theta}_I = \sum_{j \in C_I} e_j.$$

Therefore the predictor fits exactly the coordinates in C_I and misses all others. Since $\rho(A^2 - a^2)/K = 1$,

$$R_h(I) = \mathbb{E}_{(X,Y)\sim P_h}(Y - \langle \widehat{\theta}_I, X\rangle)^2 = \varepsilon\left(1 - \frac{|C_I|}{K}\right) + \mathbf{1}\{h \notin C_I\}. \tag{20}$$

Thus any subset containing the hidden coordinate has risk at most ε, whereas any subset missing it has risk at least 1.

Theorem A.1 (One point for the oracle, many points for sample-only selection)**.** *Let n be large enough that there is an integer $K \geq 8$ satisfying (A), and take the largest such K. Then $K = \Theta(n/\log n)$. For every target error $\varepsilon \in (0, 1/8)$ there is a family of well-specified noiseless linear-regression distributions $\{P_h : h \in [K]\}$ and a fixed minimum-norm learner such that the following hold.*

1. ***Full-data training is good.*** *For every $h \in [K]$, there is an event G_h with $\mathbb{P}_{S\sim P_h^n}(G_h) \geq 11/12$ such that, on G_h,*
$$R_h([n]) - F^\star = 0.$$
2. ***An evaluation oracle compresses to one example.*** *On G_h, a distribution-aware evaluation-oracle selector using at most $K - 1$ exact-risk queries outputs a subset $I_{\mathrm{or}}(S)$ of size one satisfying*
$$R_h(I_{\mathrm{or}}(S)) - F^\star \leq \varepsilon.$$
3. ***Sample-only selection needs many examples.*** *For every possibly randomized sample-only selector Π_r that outputs at most $r \leq K/4$ examples, there exists a fixed $h^\star \in [K]$ such that*
$$\mathbb{E}_{S\sim P_{h^\star}^n}\left[R_{h^\star}(I_{\Pi_r}(S)) - F^\star\right] \geq \frac{1}{2},$$
where the expectation includes the selector's internal randomness.

Consequently, to achieve expected excess risk at most ε, the evaluation oracle can select one training example, whereas every sample-only selector requires more than $K/4 = \Omega(n/\log n)$ selected examples in the worst case.

Proof. Throughout we use the risk identity (20), which we record in the dichotomous form

$$h \in C_I \implies R_h(I) \le \varepsilon, \qquad h \notin C_I \implies R_h(I) \ge 1, \tag{21}$$

where, for a subset I, $C_I \subseteq [K]$ denotes the coordinates appearing in S_I. Recall that $\widehat{\theta}_I$ fits exactly these coordinates. We write $C_{[n]}$ for the coordinates observed in the full sample, and for $j \in [K]$ we let $n_j = \#\{i \in [n] : U_i = j\}$ denote the number of training examples drawn at coordinate j.

Part 1. The full-data interpolator satisfies $C_{[n]} = [K]$ exactly when every coordinate is observed; on this event $h \in C_{[n]}$, and (21) together with the equality in (20) gives $R_h([n]) = 0 = F^\star$. Let $G_h = \{C_{[n]} = [K]\}$. By the union bound,

$$\mathbb{P}(G_h^c) \le \sum_{j=1}^{K} \mathbb{P}(j \notin C_{[n]}) = K\Big(1 - \tfrac{1}{K}\Big)^n \le K e^{-n/K}.$$

The coverage scale (A) gives $n/K \ge \log(12K)$, hence $\mathbb{P}(G_h^c) \le K e^{-\log(12K)} = 1/12$ and $\mathbb{P}(G_h) \ge 11/12$, as claimed.

Part 2. We work on G_h, where every coordinate is observed and therefore admits a singleton candidate. For $j \in [K]$, let I_j be any singleton with $C_{I_j} = \{j\}$. By the dichotomy (21), $R_h(I_h) \le \varepsilon$ while $R_h(I_j) \ge 1$ for $j \ne h$. Fix a threshold $\tau \in (\varepsilon, 1)$; the choice of τ uses knowledge of ε but not of h. The oracle queries $R_h(I_1), \dots, R_h(I_{K-1})$. If some queried value satisfies $R_h(I_j) \le \tau$, then necessarily $j = h$, and the oracle outputs I_j; otherwise every queried coordinate differs from h, so h is the single unqueried coordinate and the oracle outputs I_K. In either case the output is a singleton containing h, of risk at most ε by (21), and the oracle issues at most $K - 1$ queries.

Part 3. For the lower bound we introduce a prior $H \sim \mathrm{Unif}([K])$, used only in the analysis, and draw $S \sim P_H^n$ conditional on H. Define the event

$$E = \{C_{[n]} = [K]\} \cap \{\text{no training example has magnitude } A\}.$$

We control both constituent events by union bounds. The coordinate marginal of P_h is $\mathrm{Unif}([K])$ for every h, so the estimate of Part 1 gives $\mathbb{P}(C_{[n]} = [K]) \ge 11/12$ irrespective of H. A training example has magnitude A only when its coordinate equals H and the rare coin (probability ρ) fires, an event of probability ρ/K per example; hence $\mathbb{P}(\exists\, A\text{-example} \mid H) \le n \cdot \rho/K = 1/(16K)$. Therefore

$$\mathbb{P}(E) \ge \tfrac{11}{12} - \tfrac{1}{16K}. \tag{22}$$

The event E is $\sigma(S)$-measurable, so conditioning on E in addition to S reveals no further information. On E every observed magnitude equals a, and the coordinate pattern U has the same law under each value of H; the only dependence on H enters through the rare coin, which returns a with probability $1 - \rho$ at each of the n_H examples at the hidden coordinate and with probability 1 elsewhere. Consequently the likelihood of the realized sample under $\{H = j\}$ is proportional to $(1-\rho)^{n_j}$, and Bayes' rule with the uniform prior gives, on E,

$$\mathbb{P}(H = j \mid S) = \frac{(1-\rho)^{n_j}}{\sum_{\ell=1}^{K} (1-\rho)^{n_\ell}}. \tag{23}$$

Fix any $C \subseteq [K]$ with $|C| \le r$. Bounding each numerator term in (23) by 1 and each denominator term below by $(1-\rho)^n$ yields $\mathbb{P}(H \in C \mid S) \le r/(K(1-\rho)^n)$ on E. Since $\rho = 1/(16n)$, we have $-n\log(1-\rho) \le n\rho/(1-\rho) \le (1/16)/(15/16) = 1/15$, so $(1-\rho)^{-n} \le e^{1/15} \le e^{1/8}$ and therefore

$$\mathbb{P}(H \in C \mid S) \le e^{1/8}\, \frac{r}{K} \qquad \text{on } E. \tag{24}$$

Let Π_r be any randomized sample-only selector, with internal randomness ω independent of (H, S), and let $C(S,\omega) = C_{I_{\Pi_r}(S)}$ be the coordinate set of its output, so $|C(S,\omega)| \le r$. The dichotomy (21) gives $R_H(I_{\Pi_r}(S)) \ge \mathbf{1}\{H \notin C(S,\omega)\}$. Because ω is independent of H given S, the posterior of H given (S,ω) coincides with (23) on E, so (24) applies to $C(S,\omega)$ and, for $r \le K/4$, yields $\mathbb{P}(H \in C(S,\omega) \mid S,\omega) \le e^{1/8}/4$ on E. As $E \in \sigma(S) \subseteq \sigma(S,\omega)$, the tower rule gives

$$\mathbb{E}\big[R_H(I_{\Pi_r}(S))\big] \ge \mathbb{E}\Big[\mathbf{1}_E\big(1 - \mathbb{P}(H \in C(S,\omega) \mid S,\omega)\big)\Big] \ge \Big(1 - \frac{e^{1/8}}{4}\Big)\mathbb{P}(E).$$

Combining this with (22),

$$\mathbb{E}\big[R_H(I_{\Pi_r}(S))\big] \ge \Big(1 - \frac{e^{1/8}}{4}\Big)\Big(\frac{11}{12} - \frac{1}{16K}\Big) \ge \frac{1}{2},$$

where the final inequality uses $K \ge 8$ (the middle expression is increasing in K and exceeds 0.65 at $K = 8$). The left-hand side averages R_H over H, S, and ω; since the average over the uniform prior is at least $1/2$, there exists a fixed $h^\star \in [K]$ for which $\mathbb{E}_{S\sim P_{h^\star}^n}[R_{h^\star}(I_{\Pi_r}(S))] \ge 1/2$, which was to be shown. □

B Proof of ??

Proof. Introduce a prior $H \sim \mathrm{Unif}([K])$, used only in the analysis, and conditionally on H draw $S \sim P_H^n$ and $V \sim P_H^v$ independently. Pool the two samples into $N = n + v$ examples $Z_1, \dots, Z_N$. The selector is a measurable function of this pooled sample, the training/validation labels, and its internal randomness ω. Since the labels are independent of the coordinates and magnitudes, they carry no additional information about H beyond the observed examples. Let

$$C = C(S, V, \omega) \subseteq [K]$$

be the set of coordinates covered by the selected training examples. Then $|C| \le m$. By the risk identity (20), any output missing the hidden coordinate has excess risk at least one:

$$R_H(I_\Pi(S,V)) - F^\star \ge \mathbf{1}\{H \notin C\}. \tag{25}$$

Let $n_j = \#\{i \le N : U_i = j\}$ and let $\mu = N/K$. Define

$$E_1 = \{\text{no pooled example has magnitude } A\}, \qquad E_2 = \{\max_{\ell\in[K]} n_\ell \le 6\mu\}, \qquad E = E_1 \cap E_2.$$

A pooled example has magnitude A only when it lands on the hidden coordinate and the rare coin fires, an event of probability ρ/K. Hence

$$\mathbb{P}(E_1^c) \le \frac{N\rho}{K}.$$

Also $n_\ell \sim \mathrm{Bin}(N, 1/K)$ with mean $\mu = N/K \ge n/K \ge \log(12K)$. The multiplicative Chernoff bound gives

$$\mathbb{P}(n_\ell \ge 6\mu) \le \exp\Big(-\frac{25}{7}\mu\Big) \le (12K)^{-25/7},$$

and therefore, by a union bound,

$$\mathbb{P}(E_2^c) \le K(12K)^{-25/7} = o(1).$$

Thus

$$\mathbb{P}(E^c) \le \frac{N\rho}{K} + o(1). \tag{26}$$

On E_1 every observed magnitude equals a. The coordinate pattern has the same marginal law under every value of H, and the only dependence on H comes from the rare coin at the hidden coordinate. Thus Bayes' rule with the uniform prior gives

$$\mathbb{P}(H = j \mid S, V) = \frac{(1-\rho)^{n_j}}{\sum_{\ell=1}^{K}(1-\rho)^{n_\ell}}. \tag{27}$$

On E_2, each $n_\ell \le 6\mu$, so for every set $D \subseteq [K]$ with $|D| \le m$,

$$\mathbb{P}(H \in D \mid S, V) \le \frac{m}{K}(1-\rho)^{-6\mu} \le \frac{m}{K}\exp\!\left(\frac{6N\rho/K}{1-\rho}\right).$$

We now choose the constants in the theorem. Since K is the largest integer satisfying $n \ge K\log(12K)$, there is an absolute constant $\alpha > 0$ such that $K \ge \alpha n/\log n$ for all sufficiently large n. Choose $c_1 \le \alpha$ and $c_2 = 64$. Under the displayed hypothesis on v,

$$v \le c_1 \frac{n^2}{\log n},$$

and hence

$$\frac{N\rho}{K} = \frac{n+v}{16nK} \le \frac{1}{16K} + \frac{c_1 n}{16K\log n} \le \frac{1}{8}$$

for all sufficiently large n. Therefore the exponential factor above is bounded by an absolute constant; for definiteness,

$$\exp\!\left(\frac{6N\rho/K}{1-\rho}\right) \le 3.$$

Since ω is independent of H conditional on (S, V), conditioning further on ω does not change the posterior. Applying the last bound to the random set $C(S, V, \omega)$ gives, on E,

$$\mathbb{P}(H \in C \mid S, V, \omega) \le 3\frac{m}{K}.$$

Using (25) and the tower rule,

$$\begin{aligned}\mathbb{E}\big[R_H(I_\Pi(S,V)) - F^\star\big] &\ge \mathbb{E}\Big[\mathbf{1}_E\big(1 - \mathbb{P}(H \in C \mid S, V, \omega)\big)\Big] \\ &\ge 1 - 3\frac{m}{K} - \mathbb{P}(E^c).\end{aligned}$$

Together with (26) this yields

$$\mathbb{E}\big[R_H(I_\Pi(S,V)) - F^\star\big] \ge 1 - 3\frac{m}{K} - \frac{N\rho}{K} - o(1). \tag{28}$$

Since $N = n + v$ and $n\rho/K = 1/(16K) = o(1)$, we may rewrite this as

$$\mathbb{E}\big[R_H(I_\Pi(S,V)) - F^\star\big] \ge 1 - \frac{3m + v\rho}{K} - o(1).$$

The assumed tradeoff gives

$$v\rho = \frac{v}{16n} \le \frac{c_1 n}{16 \log n} - \frac{c_2}{16} m = \frac{c_1 n}{16 \log n} - 4m.$$

Consequently

$$3m + v\rho \le \frac{c_1 n}{16 \log n} - m \le \frac{c_1 n}{16 \log n} \le \frac{K}{16},$$

where the last inequality uses $c_1 \le \alpha$ and $K \ge \alpha n / \log n$. Substituting into the preceding lower bound,

$$\mathbb{E}\left[R_H(I_\Pi(S,V)) - F^\star\right] \ge 1 - \frac{1}{16} - o(1) \ge \frac{1}{2}$$

for all sufficiently large n.

The left-hand side averages over H, S, V, and the selector's internal randomness. Therefore some fixed $h^\star \in [K]$ satisfies

$$\mathbb{E}_{S \sim P^n_{h^\star},\, V \sim P^v_{h^\star}} \left[R_{h^\star}(I_\Pi(S,V)) - F^\star\right] \ge \frac{1}{2},$$

as claimed. □

C Proof of ??

The quadratic is structural. A hold-out point is informative about h only if it both lands on the hidden coordinate, an event of probability $1/K$, and triggers the rare high-magnitude outcome, an event of probability $\rho = 1/(16n)$. The per-point informativeness is thus ρ/K, and detecting the signal requires $\Theta(K/\rho) = \Theta(nK) = \Theta(n^2/\log n)$ draws.

Proof of Corollary **??**. Apply Theorem **??** with $m = 1$. Its condition becomes

$$v \le c_1 \frac{n^2}{\log n} - c_2 n.$$

Since $c_2 n = o(n^2/\log n)$, there is a constant $c > 0$ such that, for all sufficiently large n,

$$c \frac{n^2}{\log n} \le c_1 \frac{n^2}{\log n} - c_2 n.$$

Thus every selector using a hold-out of size $v \le cn^2/\log n$ and outputting one training example has worst-case expected excess risk at least $1/2$. On the other hand, by Theorem 3.1, on the coverage event G_h an exact-risk evaluation oracle identifies a singleton containing the hidden coordinate and hence outputs a one-example subset of excess risk at most ε. This proves the claimed quadratic validation requirement. □

Remark C.1 (The frontier is tight). The tradeoff is matched, up to constant factors, by a simple selector. A magnitude-A point can occur only at the hidden coordinate, so the decisive validation event is

$$\mathcal{B}_V := \{\exists j \le v :\ U'_j = h,\ T'_j = A\}, \qquad \mathbb{P}_h(\mathcal{B}_V) = 1 - \left(1 - \tfrac{\rho}{K}\right)^v \ \ge\ 1 - e^{-v\rho/K}.$$

Inspect V for such a point. If one is found at coordinate t, output a training example at t, after which the remaining slots are irrelevant to the risk by (20); otherwise output training examples at the m least-frequently observed coordinates. On the coverage event for S both branches are realizable, and the output

covers h unless $\mathcal{B}_V$ fails and h is not among the m least-frequent coordinates, an event of probability at most $\left(1-\frac{m}{K}\right)\left(1-\frac{\rho}{K}\right)^v$. Making this at most τ requires, up to constants, $v \gtrsim \frac{K}{\rho}\log\frac{1-m/K}{\tau}$, so for $m \ll K$ the explosion $v = \Theta(K/\rho) = \Theta(n^2/\log n)$ persists, and the output budget and validation budget trade off at the rate of Theorem ??. ◁

D Proof of the quadratic program formulation of subset selection

Proof. Recall the upper bound provided in Theorem 5.2:

$$D_{\mathrm{KL}}(\mathcal{A}_T(S) \,\|\, \mathcal{A}_T(I)) \le \frac{\beta h}{4}\sum_{k=0}^{T-1}\mathbb{E}_{\mathcal{A}_k(S)}\left[\|\nabla\Delta_I(\theta_k)\|^2\right]$$

We focus on the term $D = \sum_{k=0}^{T-1}\mathbb{E}_{\mathcal{A}_k(S)}\left[\|\nabla\Delta_I(\theta_k)\|^2\right]$: for $I \subseteq S$, let $\alpha \in \{0,1\}^n$ such that $\alpha_i = \mathbb{1}(Z_i \in I)$. Then, we have that:

$$\begin{aligned}
D &= \sum_{k=0}^{T-1}\mathbb{E}_{\mathcal{A}_k(S)}\left[\|\nabla\Delta_I(\theta_k)\|^2\right] \\
&= \sum_{k=0}^{T-1}\mathbb{E}_{\mathcal{A}_k(S)}\left[\left\|\frac{1}{m}\sum_{i=1}^{n}\alpha_i\nabla f_i(\theta_k)\right\|^2 - \frac{2}{m}\sum_{i=1}^{n}\alpha_i\langle\nabla f_i(\theta_k), \nabla F_n(\theta_k)\rangle + \|\nabla F_n(\theta_k)\|^2\right] \\
&= \sum_{k=0}^{T-1}\mathbb{E}_{\mathcal{A}_k(S)}\left[\frac{1}{m^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j\langle\nabla f_i(\theta_k), \nabla f_j(\theta_k)\rangle - \frac{2}{m}\sum_{i=1}^{n}\alpha_i\langle\nabla f_i(\theta_k), \nabla F_n(\theta_k)\rangle + \|\nabla F_n(\theta_k)\|^2\right]
\end{aligned}$$

□

E Proof of Proposition 5.1

F Proof of Theorem 5.4

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{P}^{\mathrm{SGLD}})$, where the randomness is induced by the injected noise ξ_t and the sampling of the batches B_k. Recall the two optimization trajectories induced by ULA and SGLD. Given $\theta_0 \sim \pi_0$ (both are initialized using the same distribution),

$$\theta_{k+1} = \theta_k - h\nabla L_S(\theta_k) + \sqrt{2h/\beta}\,\xi_{k+1}, \qquad \text{(ULA update)}$$
$$\theta_{k+1} = \theta_k - h\nabla L_{B_k}(\theta_k) + \sqrt{2h/\beta}\,\xi_{k+1}. \qquad \text{(SGLD update)}$$

It is established [WT11] (citation to check) that the transition kernels of the two updates are given by:

$$p^{\mathrm{ULA}}(\theta' \mid \theta) := \mathcal{N}\Big(\theta';\ \theta - h\nabla L_S(\theta),\ \tfrac{2h}{\beta}I_d\Big), \tag{29}$$
$$p^{\mathrm{SGLD}}(\theta' \mid \theta, B) := \mathcal{N}\Big(\theta';\ \theta - h\nabla \tilde{L}_B(\theta),\ \tfrac{2h}{\beta}I_d\Big). \tag{30}$$

Fix a mini-batch size $b \le n$ and let $\mathcal{B} = \{B \subseteq [n] : |B| = b\}$ with the uniform distribution $\mathbb{P}(B_k = B) = p(B) := 1/\binom{n}{b}$ for all $B \in \mathcal{B}$.

Define the filtration of the SGLD process:

$$\mathcal{G}_k := \sigma\big(\theta_0, B_0, \theta_1, B_1, \ldots, B_{k-1}, \theta_k\big), \qquad k = 0, 1, \ldots, T.$$

Under SGLD with uniform sampling, the batch B_k is drawn *after* observing θ_k and *independently* of all prior randomness, i.e.

$$B_k \perp \mathcal{G}_k \quad \text{under } \mathbb{P}^{\mathrm{SGLD}} \tag{31}$$

Definition F.1 (Per-Step and Cumulative IS Weights)**.** For $k = 0, \ldots, T-1$, define the *per-step* importance weight:

$$w_k := \frac{p^{\mathrm{ULA}}(\theta_{k+1} \mid \theta_k)}{p^{\mathrm{SGLD}}(\theta_{k+1} \mid \theta_k, B_k)}, \tag{32}$$

and the *cumulative* importance weight for step k:

$$W_{0:k} := \prod_{t=0}^{k-1} w_t, \qquad W_{0:0} := 1. \tag{33}$$

Definition F.2 (SGLD Gram Matrix Estimator)**.** Let π_{ij} be the probability that both Z_i and Z_j appear in a uniformly drawn batch of size b:

$$\pi_{ij} = \begin{cases} \dfrac{b(b-1)}{n(n-1)} & i \neq j, \\ \dfrac{b}{n} & i = j. \end{cases}$$

The *importance sampling estimator* of P_{ij}^{ULA} is:

$$\hat{P}_{ij}^{\mathrm{SGLD}} := \sum_{k=0}^{T-1} W_{0:k} \cdot \frac{\mathbb{1}[i \in B_k,\ j \in B_k]}{\pi_{ij}} \big\langle \nabla \ell(\theta_k, Z_i),\ \nabla \ell(\theta_k, Z_j) \big\rangle. \tag{34}$$

We first derive a closed form for w_k. We have:

$$\begin{aligned} w_k &:= \frac{p^{\mathrm{ULA}}(\theta_{k+1} \mid \theta_k)}{p^{\mathrm{SGLD}}(\theta_{k+1} \mid \theta_k, B_k)} \\ &= \exp\left(\frac{\beta}{4h} \left(\left\| \theta_{k+1} - \theta_k + h \nabla \tilde{L}_{B_k}(\theta_k) \right\|^2 - \left\| \theta_{k+1} - \theta_k + h \nabla L_S(\theta_k) \right\|^2 \right) \right). \end{aligned} \tag{35}$$

Writing $r = \theta_{k+1} - \theta_k$, we have that:

$$\|\mathbf{r} + h \nabla \tilde{L}_{B_k}\|^2 - \|\mathbf{r} + h \nabla L_S\|^2 = 2h \langle \mathbf{r},\ \nabla \tilde{L}_{B_k} - \nabla L_S \rangle + h^2 \big(\|\nabla \tilde{L}_{B_k}\|^2 - \|\nabla L_S\|^2 \big).$$

Thus, substituting back into Equation (35) yields:

$$w_k = \exp\left(\frac{\beta}{2} \big\langle \theta_{k+1} - \theta_k,\ \nabla \tilde{L}_{B_k}(\theta_k) - \nabla L_S(\theta_k) \big\rangle + \frac{\beta h}{4} \left(\left\| \nabla \tilde{L}_{B_k}(\theta_k) \right\|^2 - \left\| \nabla L_S(\theta_k) \right\|^2 \right) \right). \tag{36}$$

Lemma F.3 (Marginal Change of Measure)**.** *For every $k \in \{0, 1, \ldots, T-1\}$ and every bounded measurable function $f : \mathbb{R}^d \to \mathbb{R}$:*

$$\mathbb{E}^{\mathrm{SGLD}}\big[W_{0:k} \cdot f(\theta_k) \big] = \mathbb{E}^{\mathrm{ULA}}\big[f(\theta_k) \big]. \tag{37}$$

Proof. We proceed by induction on k.
k = 0.:

$W_{0:0} = 1$ by definition, and both chains share the initialisation $\theta_0 \sim \mu_0$, so:

$$\mathbb{E}^{\mathrm{SGLD}}\big[1 \cdot f(\theta_0)\big] = \int_{\mathbb{R}^d} f(\theta)\, \mu_0(d\theta) = \mathbb{E}^{\mathrm{ULA}}\big[f(\theta_0)\big].$$

Suppose (37) holds for step k. We prove it for step $k+1$. Noting that $W_{0:k+1} = W_{0:k} \cdot w_k$: Noting that $W_{0:k+1} = W_{0:k} \cdot w_k$, using the tower property yields:

$$\mathbb{E}^{\mathrm{SGLD}}_{\mathcal{G}_{k+1}}\big[W_{0:k+1} \cdot f(\theta_{k+1})\big] = \mathbb{E}^{\mathrm{SGLD}}_{\mathcal{G}_k}\Bigg[W_{0:k} \cdot \underbrace{\mathbb{E}^{\mathrm{SGLD}}_{B_k,\theta_{k+1}}\left[\frac{p^{\mathrm{ULA}}(\theta_{k+1} \mid \theta_k)}{p^{\mathrm{SGLD}}(\theta_{k+1} \mid \theta_k, B_k)} \cdot f(\theta_{k+1}) \,\middle|\, \mathcal{G}_k\right]}_{=:\,\Phi(\theta_k)}\Bigg], \tag{38}$$

Conditioned on $\mathcal{G}_k$ (equivalently, on θ_k), the pair (B_k, θ_{k+1}) is generated under $\mathbb{P}^{\mathrm{SGLD}}$ as:

$$\mathbb{P}((B_k, \theta_{k+1}) \mid \theta_k) = \mathbb{P}(B_k) \cdot \mathbb{P}^{\mathrm{SGLD}}(\theta_{k+1} \mid \theta_k, B_k). \tag{39}$$

Expanding the conditional expectation using (39):

$$\begin{aligned}
\Phi(\theta_k) &= \sum_{B \in \mathcal{B}} p(B) \int_{\mathbb{R}^d} p^{\mathrm{SGLD}}(\theta' \mid \theta_k, B) \cdot \frac{p^{\mathrm{ULA}}(\theta' \mid \theta_k)}{p^{\mathrm{SGLD}}(\theta' \mid \theta_k, B)} \cdot f(\theta')\, d\theta' \\
&= \sum_{B \in \mathcal{B}} p(B) \int_{\mathbb{R}^d} \frac{p^{\mathrm{SGLD}}(\theta' \mid \theta_k, B)}{p^{\mathrm{SGLD}}(\theta' \mid \theta_k, B)} p^{\mathrm{ULA}}(\theta' \mid \theta_k) \cdot f(\theta')\, d\theta', \qquad (40) \\
&= \Big(\sum_{B \in \mathcal{B}} p(B)\Big) \cdot \int_{\mathbb{R}^d} p^{\mathrm{ULA}}(\theta' \mid \theta_k) \cdot f(\theta')\, d\theta', \\
&= \int_{\mathbb{R}^d} p^{\mathrm{ULA}}(\theta' \mid \theta_k) f(\theta')\, d\theta' \;=\; \mathbb{E}^{\mathrm{ULA}}\big[f(\theta_{k+1}) \mid \theta_k\big], \qquad (41)
\end{aligned}$$

Thus, $\Phi(\theta_k) = \mathbb{E}^{\mathrm{ULA}}[f(\theta_{k+1}) \mid \theta_k]$, which is a bounded measurable function of θ_k. Substituting (41) into (38):

$$\mathbb{E}^{\mathrm{SGLD}}_{\mathcal{G}_{k+1}}\big[W_{0:k+1} \cdot f(\theta_{k+1})\big] = \mathbb{E}^{\mathrm{SGLD}}_{\mathcal{G}_k}\big[W_{0:k} \cdot \Phi(\theta_k)\big].$$

Applying the inductive hypothesis (37) to the bounded measurable function

$$\mathbb{E}^{\mathrm{SGLD}}_{\mathcal{G}_k}\big[W_{0:k} \cdot \Phi(\theta_k)\big] \;=\; \mathbb{E}^{\mathrm{ULA}}_{\theta_{0:k}}\big[\Phi(\theta_k)\big] \;=\; \mathbb{E}^{\mathrm{ULA}}_{\theta_{0:k}}\Big[\mathbb{E}^{\mathrm{ULA}}_{\theta_{k+1}}\big[f(\theta_{k+1}) \mid \theta_k\big]\Big] \;=\; \mathbb{E}^{\mathrm{ULA}}_{\theta_{0:k+1}}\big[f(\theta_{k+1})\big],$$

where the last equality is the tower property under $\mathbb{P}^{\mathrm{ULA}}$. This completes the induction. □

Theorem 5.4 (Unbiasedness of the Path IS Estimator)**.** *For all $i, j \in [n]$:*

$$\mathbb{E}^{\mathrm{SGLD}}\Big[\hat{P}^{\mathrm{path}}_{ij}\Big] = P^{\mathrm{ULA}}_{ij} \;:=\; \sum_{k=0}^{T-1} \mathbb{E}_{\theta_k \sim \mathcal{A}_k(S)}\big[\big\langle \nabla \ell(\theta_k, Z_i),\ \nabla \ell(\theta_k, Z_j) \big\rangle\big].$$

Proof. By linearity of expectation and definition (34):

$$\mathbb{E}^{\mathrm{SGLD}}\Big[\hat{P}_{ij}^{\mathrm{path}}\Big] = \sum_{k=0}^{T-1} \mathbb{E}^{\mathrm{SGLD}}\left[W_{0:k} \cdot \frac{\mathbb{1}[i \in B_k,\ j \in B_k]}{\pi_{ij}} \big\langle g_i(\theta_k),\, g_j(\theta_k)\big\rangle\right], \tag{42}$$

where we write $g_\ell(\theta) := \nabla \ell(\theta, Z_\ell)$ for brevity.

Conditioning on $\mathcal{G}_k$ and exploiting the independence (31) ($B_k \perp \mathcal{G}_k$):

$$\begin{aligned}
&\mathbb{E}^{\mathrm{SGLD}}\left[W_{0:k} \cdot \frac{\mathbb{1}[i \in B_k,\ j \in B_k]}{\pi_{ij}} \big\langle g_i(\theta_k),\, g_j(\theta_k)\big\rangle\right] \\
&\quad = \mathbb{E}^{\mathrm{SGLD}}\Bigg[W_{0:k} \cdot \langle g_i(\theta_k), g_j(\theta_k)\rangle \cdot \underbrace{\mathbb{E}^{\mathrm{SGLD}}\left[\frac{\mathbb{1}[i \in B_k,\ j \in B_k]}{\pi_{ij}} \,\middle|\, \mathcal{G}_k\right]}_{(\star)}\Bigg].
\end{aligned} \tag{43}$$

Since $B_k \perp \mathcal{G}_k$, the conditional expectation $(\star)$ equals the unconditional expectation:

$$(\star) \;=\; \frac{\mathbb{E}[\mathbb{1}[i \in B_k,\ j \in B_k]]}{\pi_{ij}} \;=\; \frac{\pi_{ij}}{\pi_{ij}} \;=\; 1.$$

Substituting back into (43):

$$\mathbb{E}^{\mathrm{SGLD}}\left[W_{0:k} \cdot \frac{\mathbb{1}[i \in B_k,\ j \in B_k]}{\pi_{ij}} \big\langle g_i(\theta_k),\, g_j(\theta_k)\big\rangle\right] = \mathbb{E}^{\mathrm{SGLD}}\big[W_{0:k} \cdot \langle g_i(\theta_k),\, g_j(\theta_k)\rangle\big]. \tag{44}$$

The function $f(\theta) := \langle g_i(\theta), g_j(\theta)\rangle$ is bounded and measurable. Applying Lemma F.3 to the right-hand side of (44):

$$\mathbb{E}^{\mathrm{SGLD}}\big[W_{0:k} \cdot \langle g_i(\theta_k),\, g_j(\theta_k)\rangle\big] \;=\; \mathbb{E}^{\mathrm{ULA}}\big[\langle g_i(\theta_k),\, g_j(\theta_k)\rangle\big]. \tag{45}$$

Combining (42), (44), and (45):

$$\mathbb{E}^{\mathrm{SGLD}}\Big[\hat{P}_{ij}^{\mathrm{path}}\Big] = \sum_{k=0}^{T-1} \mathbb{E}^{\mathrm{ULA}}\big[\langle g_i(\theta_k), g_j(\theta_k)\rangle\big] = P_{ij}^{\mathrm{ULA}}. \quad \square$$